STAT 400 Homework 6: Random Variables

1) **Parking Enforcement (Poisson Distribution)**
Suppose parking enforcement visits the lot according to the Poisson distribution with a rate $\lambda = 0.5$ visits per hour. When they visit, all cars without a permit get ticketed. The paid parking hours are 8:00 AM to 4:00 PM (8 hours total).

a) What is the probability that parking enforcement visits the lot at least once during the 8-hour day?
(Hint: Calculate the probability of 0 visits and subtract from 1).

$$P(X \geq 1) = 1 - P(X = 0)$$
$$P(X \geq 1) = 1 - e^{-8\lambda}\frac{\lambda^0}{0!}$$
$$P(X \geq 1) = 1 - e^{-4}$$
$$P(X \geq 1) = \mathbf{0.982}$$

b) What is the probability that John can park illegally for 10 days until he gets his first ticket? State any assumptions you are making.

I will assume that John's car will be in the parking lot for the entire paid duration (8:00 AM to 4:00 PM).

$$P(X = 0) = e^{-80\lambda}\frac{\lambda^0}{0!}$$
$$P(X = 0) = \boldsymbol{e^{-40}}$$

c) What is the expected number of days John can park illegally before he gets a ticket?

Let Y denote the number of days before John gets a ticket.

$$\therefore Y \sim \text{Geometric}\left(1 - e^{-4}\right)$$

$$E(Y) = \frac{1}{1 - e^{-4}}$$
$$E(Y) = \mathbf{1.019\ days}$$

1) d) Find an estimate for the number of days, d, such that the probability of waiting d days or more for the first ticket is 0.975.
(Hint: Use the Empirical Rule).

$$
\begin{aligned}
P(Y \geq d) &= 0.975 \\
(1-p)^{d-1} &= 0.975 \\
(1-(1-e^{-}4))^{d-1} &= 0.975 \\
\left(e^{-4}\right)^{d-1} &= 0.975 \\
e^{4-4d} &= 0.975 \\
4-4d &= \ln(0.975) \\
d &= \frac{\ln(0.975)-4}{-4} \\
d &= \mathbf{1.006\ days}
\end{aligned}
$$

2) **Casino Roulette (Geometric Distribution)**
At a casino, the number of trials needed to win a round of roulette follows a Geometric distribution with a success probability of $p = 0.2$. Let X be the trial number of your first win.

a) Find the probability that you win a round:

i. in the first two trials?

$$P(X \leq 2) = P(X = 1) + P(X = 2)$$
$$P(X \leq 2) = p + (1 - p)p$$
$$P(X \leq 2) = 0.2 + (0.8)0.2$$
$$P(X \leq 2) = \mathbf{0.36}$$

ii. within the first 10 trials?

$$P(X \leq 10) = 1 - P(X > 10)$$
$$P(X \leq 10) = 1 - (1 - p)^{10}$$
$$P(X \leq 10) = 1 - (0.8)^{10}$$
$$P(X \leq 10) = \mathbf{0.893}$$

iii. in more than 3 trials?

$$P(X > 3) = (1 - p)^3$$
$$P(X > 3) = (0.8)^3$$
$$P(X > 3) = \mathbf{0.512}$$

iv. between 10 and 20 trials?

$$P(10 < X < 20) = P(11 \leq X \leq 19)$$
$$P(10 < X < 20) = P(X \leq 19) - P(X \leq 11)$$
$$P(10 < X < 20) = 1 - P(X > 19) - (1 - P(X > 11))$$
$$P(10 < X < 20) = P(X > 11) - P(X > 19)$$
$$P(10 < X < 20) = (1 - p)^{11} - (1 - p)^{19}$$
$$P(10 < X < 20) = (0.8)^{11} - (0.8)^{19}$$
$$\boldsymbol{P(10 < X < 20) = 0.0715}$$

b) Suppose we track 10 independently selected people. We deem a person a "success" if they win a round within 3 trials. Let Y track the number of successful people out of the 10.

i. What is the distribution of Y? (State the name and parameters).

$$Y \sim \text{Binomial}\left(10, 1 - 0.8^3\right)$$
$$\boldsymbol{Y \sim \textbf{Binomial}(10, 0.488)}$$

2) **ii.** What is the probability that there are at least two successful people?

$$P(2 \leq Y \leq 10) = \sum_{y=2}^{10} \binom{10}{y} p^y (1-p)^{10-y}$$

$$P(2 \leq Y \leq 10) = \sum_{y=2}^{10} \binom{10}{y} 0.488^y (0.512)^{10-y}$$

$$P(2 \leq Y \leq 10) = \mathbf{0.987}$$

iii. What is the expected number of successful people out of 10?

$$E(Y) = 10p$$

$$E(Y) = 10(0.488)$$

$$E(Y) = \mathbf{4.88\ people}$$

3) **Car Wash (Exponential Distribution)**
Suppose X models the time it takes (in hours) to wash a car. The probability density function (PDF) is:

$$f_X(x) = 2e^{-2x} \quad \text{for } x > 0$$

(Note: This is an Exponential distribution with $\lambda = 2$. The mean of an Exponential distribution is $1/\lambda$).

a) What is the average amount of time it takes to wash a car?

$$E(X) = \mathbf{\frac{1}{2}\ hours}$$

b) A car is deemed a "success" if it takes less than the average time to wash. Calculate $P(\text{Success})$.

$$P(\text{Success}) = P\left(0 < X < \frac{1}{2}\right)$$

$$P(\text{Success}) = \int_0^{\frac{1}{2}} 2e^{-2x}\,\mathrm{d}x$$

$$P(\text{Success}) = \left(-e^{-2x}\right)\Big|_0^{\frac{1}{2}}$$

$$P(\text{Success}) = 1 - e^{-1}$$

$$P(\text{Success}) = \mathbf{0.632}$$

c) Among 15 randomly arriving cars, let Y be the number of successes. What is the distribution of Y? What is the expected number of successes?

$$Y \sim \mathbf{Binom}\left(\mathbf{15}, \mathbf{1 - e^{-1}}\right)$$

$$E(Y) = 15p$$

$$E(Y) = 15\left(1 - e^{-1}\right)$$

$$E(Y) = \mathbf{9.482\ successes}$$

d) Suppose we keep track of cars. What is the expected number of cars we need to track until we get a successful car?

Let Z denote the number of cars before a successful car.

$$\therefore Z \sim \text{Geometric}\left(1 - e^{-1}\right)$$

$$E(Z) = \frac{1}{p}$$

$$E(Z) = \frac{1}{1 - e^{-1}}$$

$$E(Z) = \mathbf{1.582\ cars}$$